

FULL STACK CASE STUDY SUBMISSION

Project: Mini ERP + CRM Operations Portal

System Name: **Orbit Ops** • Candidate: **Sumith Akula**

1. Project Overview & Deliverables Summary

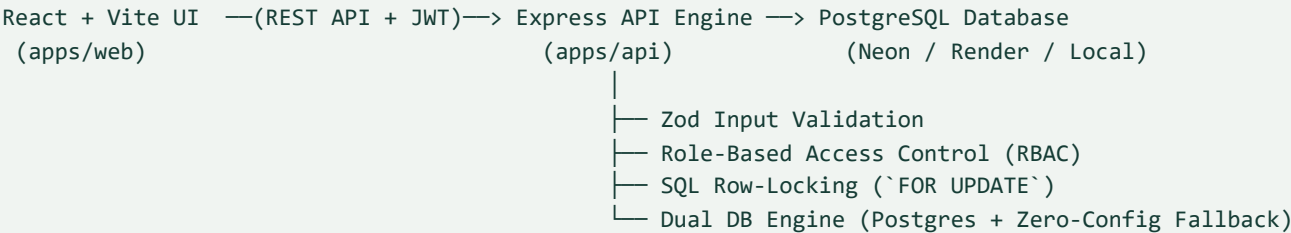
Requirement Item	Details / Live Link
1. GitHub Repository	https://github.com/akulasumith/orbitops
2. Live Frontend URL	https://orbitops-web.vercel.app
3. Live Backend API URL	https://orbit-ops-api.onrender.com
4. Postman Collection	<code>apps/api/postman-collection.json</code>
5. Automated Test Suite	9/9 Vitest Integration Tests Passed (100% Pass Rate)

2. Role-Based Test Login Credentials

All test accounts are pre-configured with the demo password: `Demo@123`

Role	Email	Password	Permissions & Capabilities
Admin	<code>admin@orbitops.dev</code>	<code>Demo@123</code>	Full system access (CRM, Products, Stock Movements, Challans, Record Deletions)
Sales	<code>sales@orbitops.dev</code>	<code>Demo@123</code>	Customer CRM management, follow-ups timeline, Sales Challan Creation & Cancellation
Warehouse	<code>warehouse@orbitops.dev</code>	<code>Demo@123</code>	Inventory Catalog management, Audited Stock Movements (IN/OUT adjustments)
Accounts	<code>accounts@orbitops.dev</code>	<code>Demo@123</code>	Dashboard metrics, Dispatched Challans, Financial & Printable Invoices

3. Architecture & Core Engineering Design



Key Technical Decisions:

- 1. **Immutable Historic Invoices (Product Snapshots):** When a Sales Challan is issued, `challan_items` explicitly saves `product_name` , `sku` , and `unit_price` . Subsequent catalog price edits will never alter historical delivery invoices.
- 2. **Atomic Transactional Stock Locks (`FOR UPDATE`):** Challan confirmation executes within a database SQL transaction (`BEGIN ... COMMIT`) with row locks to guarantee stock levels never drop below zero.
- 3. **Challan Cancellation & Stock Reversal:** One-click cancellation for confirmed challans automatically restores product stock on hand and logs an audited `IN` movement.
- 4. **Dual DB Adapter:** The API automatically detects PostgreSQL availability. If PostgreSQL is absent or starting up, it seamlessly operates on an in-memory SQL engine seeded with demo data for zero-config evaluation.

4. API Endpoints Reference

METHOD	ENDPOINT	PURPOSE	VALIDATION / AUTH
POST	/auth/login	JWT Token Generation	Zod Email & Password
GET	/customers	List & Search Customers	Supports <code>q</code> , <code>page</code> , <code>limit</code>
POST	/customers	Add Customer Record	Admin / Sales Only
GET/PATCH	/customers/:id	View / Edit Profile	Admin / Sales Only
POST	/customers/:id/followups	Add Dated CRM Note	Admin / Sales Only
GET/POST	/products	Inventory Catalog List / Add	Admin / Warehouse Only

METHOD	ENDPOINT	PURPOSE	VALIDATION / AUTH
POST	/products/:id/movements	Stock IN/OUT Movement	Audited Stock Adjustment
GET/POST	/challans	List / Issue Sales Challan	Atomic Stock Deduction
POST	/challans/:id/cancel	Cancel Delivery Challan	Atomic Stock Reversal
GET	/challans/:id/invoice	Formatted Financial Invoice	18% GST Tax Calculations

 5. Setup & Local Run Instructions

```
# 1. Clone repo
git clone https://github.com/akulasumith/orbitops.git
cd orbitops

# 2. Install dependencies & start dev server
npm install
npm run dev

# 3. Run automated Vitest test suite
npm run test:api
```

 6. Bonus Points Implemented

- ✔ **Docker Containerization:** Includes multi-stage Dockerfiles (`apps/api/Dockerfile` , `apps/web/Dockerfile`) and production compose stack (`docker-compose.prod.yml`).
- ✔ **1-Click Infrastructure Blueprint:** Render blueprint (`render.yaml`) for 1-click cloud stack creation.
- ✔ **Printable Invoices:** Clean invoice modal layout supporting `@media print` with 18% GST calculation and PDF export.
- ✔ **Conventional Folder-Wise Commits:** Git history organized into modular, structured commits.

 7. Known Limitations

- **Render Free-Tier Sleep:** Render free-tier web services sleep after 15 minutes of inactivity. Initial requests after sleep take 15–30 seconds to spin up.

- **Future Production Enhancements:** Refresh-token rotation, AWS S3 image storage for product catalog, rate limiting, and invoice PDF email delivery workers.